RES-5 / GAP-2 entry: the matched-order-to-exact remainder is common-mode

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-09 · this version issued 2026-06-09 · v1.0

1. Purpose and scope

After the RES-1 off-diagonal arc, the H-LAYER residual consolidated to RES-5 (GAP-2): converting the matched-second-cumulant selection $F^{(2)}[\mathcal{P}] > F^{(2)}[\mathcal{R}_H]$ into a statement about EXACT free energies. The operator directed a direct attack on RES-5 (the exchange-Hessian eigenvalue + exact remainder, unified). This note opens it with the common-mode mechanism: the matched-order structure makes the leading beyond-second-cumulant correction CANCEL in the selection difference, leaving a remainder controlled by the condensate fraction a_0. Scope: the leading common-mode reduction and the control parameter; the all-order structure is the residual.

2. The matched-order-to-exact problem

$$\Delta F_{\text{exact}} = \Delta F^{(2)} + \sum_{n\geq 3} \Delta F^{(n)}, \qquad \Delta F^{(n)} = F^{(n)}[\mathcal{P}] - F^{(n)}[\mathcal{R}_H].$$

RES-5 closes iff $\left|\sum_{n\geq 3} \Delta F^{(n)}\right| < \Delta F^{(2)}$ (the GAP-2 controlled-error condition $\Delta F^{(2)} - \varepsilon_{\text{ctrl}} > 0$).

3. The common-mode mechanism (extending R-U10-3 to the bulk)

At fixed intensity I the diagonal dressing

$$\hat{r}(I) = r_R + 2\lambda' I$$

is PATTERN-INDEPENDENT: $\mathcal{P}$ and $\mathcal{R}_H$ share the same dressed inverse propagator $D_0(I)$. This is exactly the structure the near-gap resolution (R-U10-3, neargap-common-mode-resolution) used at second cumulant to prove $\Delta F^{(2)} \geq 0$ unconditionally. Extending it: each beyond-second-cumulant correction $F^{(n)}$ splits into

$$F^{(n)} = \underbrace{F^{(n)}_{\text{cm}}(D_0)}_{\text{common-mode, pattern-independent}} + \underbrace{F^{(n)}_{\text{pd}}(D_0; \mathcal{P}^2)}_{\text{pattern-dependent}},$$

and the common-mode part CANCELS in $\Delta F^{(n)} = F^{(n)}[\mathcal{P}] - F^{(n)}[\mathcal{R}_H]$. The pattern-dependent part is carried by the condensate's relative weight in the dressing,

$$a_0(I) = \frac{2\lambda' I}{\hat{r}(I)},$$

so $\Delta F^{(n)} = O(a_0)\, F^{(n)}$: the matched-order difference is $O(a_0)$ suppressed relative to the individual cumulants.

4. The leading envelope at the operating intensity

$$\boxed{a_0(I) = \begin{cases} 0.021, & I = 4\times10^{-4} \\ 0.050, & I = 10^{-3} \\ 0.096, & I = 2\times10^{-3}\ \text{(endpoint)} \end{cases} \qquad \Delta F_{\text{exact}} \ \gtrsim\ \Delta F^{(2)}(1-a_0) \ \geq\ 0.904\,\Delta F^{(2)} > 0.}$$

The condensate carries only $\sim 10\%$ of the dressing at the operating endpoint, so the leading beyond-second-cumulant remainder is $\lesssim 10\%$ of the second-cumulant margin: the selection survives the matched-order-to-exact correction at leading common-mode order. a_0 is monotone in I, worst at the $I = 2\times10^{-3}$ endpoint – consistent with the RES-4 / SC-SCOPE endpoint being the thinnest.

5. Consistency with the established partial results

The a_0-envelope unifies the prior RES-5-adjacent results: ESTIMATOR-UPGRADE (CLOSED@controlled-error for the margins) is the error-bound realisation; SC-SCOPE (third-cumulant endpoint, thin-certified $\times1.04$) is the explicit $n = 3$ term, consistent with a $\sim 10\%$ correction; the exchange-Hessian (RES-1b, condensate-direction $R_{\text{lead}} \leq 0.174$) is an off-diagonal beyond-second contribution, also $O(a_0)$-scaled. The common-mode envelope is the single control parameter (a_0) organising all three.

6. Status and residual

T3 PROOF SKETCH. Established: the common-mode control parameter $a_0 \leq 0.096$ and the leading envelope $\Delta F_{\text{exact}} \gtrsim \Delta F^{(2)}(1-a_0) > 0$. **OPEN (the deep RES-5 content):** (i) the ALL-ORDER common-mode structure (that every $F^{(n)}$ splits with the pattern-dependent part $O(a_0)$, not just $n = 3$); (ii) the precise coefficient (is the remainder exactly $a_0\Delta F^{(2)}$ or $C\,a_0\Delta F^{(2)}$ with C bounded?); (iii) the geometric / convergence bound on $\sum_{n\geq3}$. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. This is the entry to RES-5, not its closure.

7. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"You show the leading ($n = 3$) common-mode, then assume all orders."* VALID and stated: the $\hat{r}$ pattern-independence is rigorous and gives the leading suppression; the all-order common-mode split is the named residual (i). The note is the entry, T3, not a closure.

(β) *"$O(a_0)$ hides the coefficient – it could be Ca_0 with $C > 10$."* VALID-with-mitigation: the structural a_0 scaling is the result; the constant C requires the explicit cumulant-difference computation (residual ii). The SC-SCOPE $n = 3$ value ($\times1.04$, a $\sim 4\%$ effect) is consistent with $C = O(1)$, but is not a proof of $C = O(1)$ at all orders.

(γ) *"This re-treads SC-SCOPE / ESTIMATOR-UPGRADE."* PARTLY: it UNIFIES them under the single common-mode control parameter a_0 (§5), which is the new organising principle; it does not re-derive them.

Quantitative sanity check (mandatory): *magnitude* – $a_0(2e{-}3) = 0.096$, so $(1-a_0) = 0.904 > 0$; *monotone* – $a_0 = 0.021/0.050/0.096$ increasing in I (endpoint worst); *consistency* – the $n = 3$ SC-SCOPE correction ($\sim 4\%$) sits within the $a_0 \sim 10\%$ envelope; *structure* – $\hat{r}(I)$ depends only on

I, not the pattern, so the common-mode cancellation is exact at the dressing level (the R-U10-3 mechanism).

8. Result footer

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Result ID:               B1-RH-ENUM / RES-5 GAP-2 entry (common-mode envelope)
Precise statement:       The matched-order-to-exact remainder sum_{n>=3} Delta
                         F^(n) is common-mode suppressed: r_hat(I)=rR+2 lam' I is
                         pattern-independent, so the common-mode part cancels in
                         the difference and the residual is O(a0), a0=2 lam' I/
                         r_hat = 0.021/0.050/0.096 at I=4e-4/1e-3/2e-3. Hence
                         Delta F_exact >~ Delta F^(2)(1-a0) >= 0.904 Delta F^(2)>0
                         at the operating endpoint, at leading common-mode order.
Scope:                   Leading (n=3) common-mode reduction + the a0 control
                         parameter; anchor mu^2, operating interval. All-order
                         common-mode + the coefficient bound are OPEN.
Dependencies:            neargap-common-mode-resolution (R-U10-3, the 2nd-cumulant
                         mechanism); SC-SCOPE (n=3 endpoint); ESTIMATOR-UPGRADE
                         (controlled-error margins); exchange-Hessian (RES-1b);
                         Math424-AddA (rR, M_R, lam').
Evidence grade:          ANALYTIC (common-mode structure) + EXECUTED
                         (res5_commonmode_envelope.py 4/4).
Reproduction command:    python codes/vacuum/res5_commonmode_envelope.py
Expected output:         a0 = 0.021/0.050/0.096; (1-a0) = 0.979/0.950/0.904; 4/4.
Falsification gate:      a0 >= 1 at operating I; or a beyond-2nd-cumulant
                         difference NOT O(a0) (a pattern-dependent term surviving
                         the common-mode cancellation at O(1)); or the SC-SCOPE
                         n=3 value exceeding the a0 envelope.
Tier before / after:     No tier flip. B1 T6 on {H-LAYER} unchanged. RES-5/GAP-2:
                         entry opened; leading common-mode envelope a0<=0.096.
No-overclaim statement:  This does NOT close RES-5/GAP-2. Only the LEADING
                         common-mode suppression and the a0 control parameter are
                         established; the all-order common-mode split, the
                         coefficient C, and the cumulant-series convergence are
                         the deep residual. B1 stays T6 on {H-LAYER}.
Next required action:    Prove the all-order common-mode split (every F^(n)'s
                         pattern-dependent part is O(a0)) and bound the coefficient
                         C, OR a direct geometric bound on sum_{n>=3} Delta F^(n);
                         then RES-5 -> controlled-error theorem.
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